PYTHON PROGRAMS

AIM • PROCEDURE • FLOWCHART • PROGRAM • OUTPUT

Practical Record

Note: The outputs shown are sample runs using the sample input values printed in each section. Actual output changes according to the values entered by the user.

PROGRAM 1: Salary Calculation and Deduction

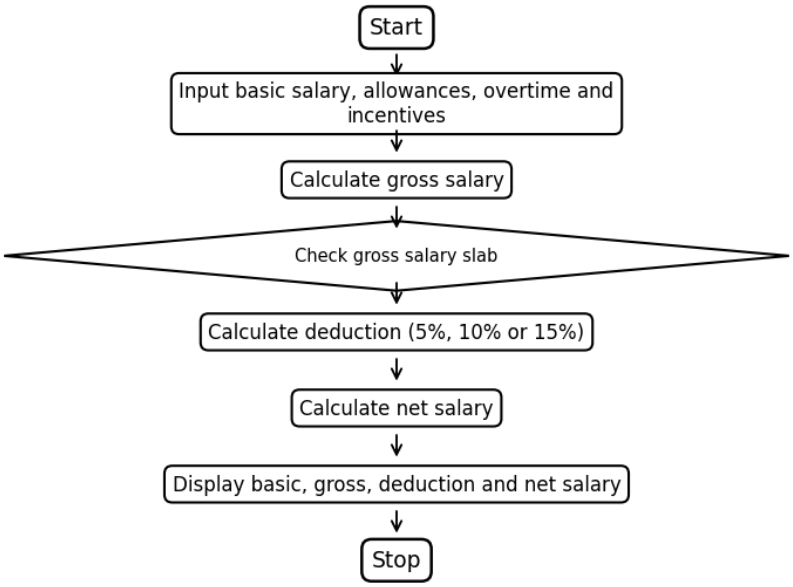
AIM

Calculate gross salary from basic salary, allowances, overtime and incentives. Apply deductions based on the gross salary slab and find net salary.

PROCEDURE

1. Input basic salary, allowances, overtime and incentives
2. Calculate gross salary
3. Check gross salary slab
4. Calculate deduction (5%, 10% or 15%)
5. Calculate net salary
6. Display basic, gross, deduction and net salary

FLOWCHART



PROGRAM

```
basic = float(input("Enter basic salary: "))
allowances = float(input("Enter allowances: "))
overtime = float(input("Enter overtime amount: "))
incentives = float(input("Enter incentives: "))

gross = basic + allowances + overtime + incentives

if gross <= 20000:
    deduction = gross * 0.05
elif gross <= 40000:
    deduction = gross * 0.10
else:
    deduction = gross * 0.15

net = gross - deduction

print("basic Salary:", basic)
print("gross Salary:", gross)
print("deduction:", deduction)
print("net Salary:", net)
```

OUTPUT (SAMPLE)

```
Enter basic salary: 15000
Enter allowances: 3000
Enter overtime amount: 1000
Enter incentives: 1000
basic Salary: 15000.0
gross Salary: 20000.0
deduction: 1000.0
net Salary: 19000.0
```

PROGRAM 2: Mobile Data Plan Eligibility

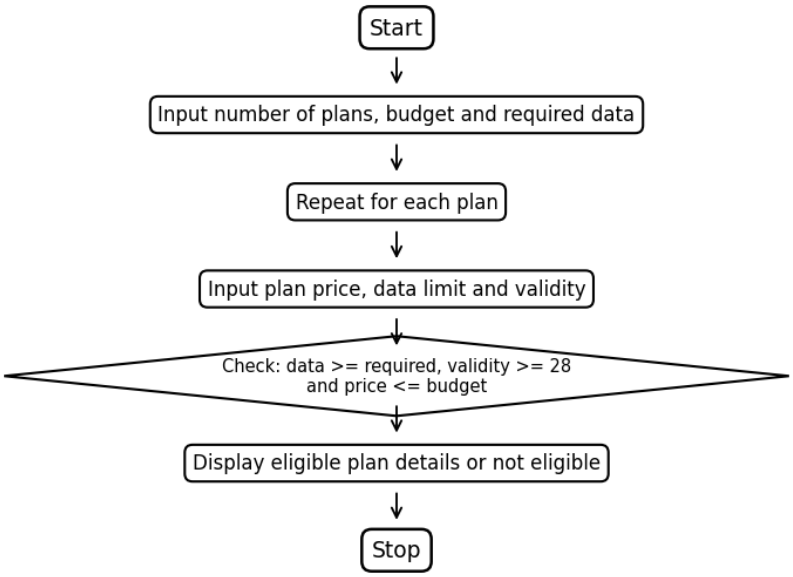
AIM

Check each mobile plan against the user's budget, required data and minimum validity of 28 days.

PROCEDURE

1. Input number of plans, budget and required data
2. Repeat for each plan
3. Input plan price, data limit and validity
4. Check: data >= required, validity >= 28 and price <= budget
5. Display eligible plan details or not eligible

FLOWCHART



PROGRAM

```
n = int(input("Enter number of plans: "))
budget = int(input("Enter your budget: "))
usage = int(input("Enter required data in GB: "))

for i in range(n):
    print("Plan", i + 1)

    price = int(input("Enter plan price: "))
    data = int(input("Enter data limit in GB: "))
    validity = int(input("Enter validity in days: "))

    if data >= usage and validity >= 28 and price <= budget:
        print("eligible plan")
        print("plan price:", price)
    else:
        print("not eligible")
```

OUTPUT (SAMPLE)

```
Enter number of plans: 2
Enter your budget: 300
Enter required data in GB: 2
Plan 1
Enter plan price: 249
Enter data limit in GB: 2
Enter validity in days: 28
eligible plan
plan price: 249
Plan 2
Enter plan price: 399
Enter data limit in GB: 3
Enter validity in days: 56
not eligible
```

PROGRAM 3: Vehicle Fuel Consumption and Route Cost

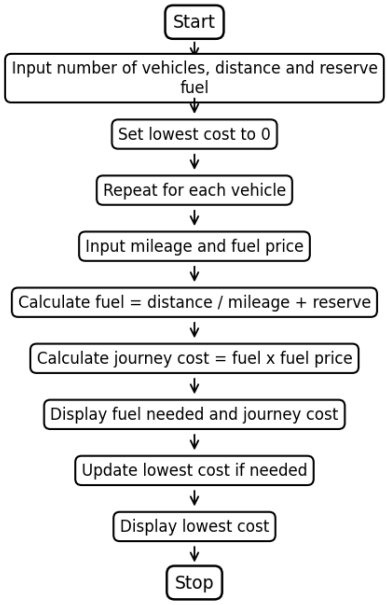
AIM

Calculate fuel required and journey cost for each vehicle, including reserve fuel, and identify the lowest journey cost.

PROCEDURE

1. Input number of vehicles, distance and reserve fuel
2. Set lowest cost to 0
3. Repeat for each vehicle
4. Input mileage and fuel price
5. Calculate fuel = distance / mileage + reserve
6. Calculate journey cost = fuel x fuel price
7. Display fuel needed and journey cost
8. Update lowest cost if needed
9. Display lowest cost

FLOWCHART



PROGRAM

```
n = int(input("Enter number of vehicles: "))
distance = int(input("Enter distance: "))
reserve = int(input("Enter reserve fuel: "))

low = 0

for i in range(n):
    mileage = int(input("Enter mileage: "))
    price = int(input("Enter fuel price: "))

    fuel = distance / mileage
    fuel = fuel + reserve
    cost = fuel * price

    print("fuel needed:", fuel)
    print("journey cost:", cost)

    if low == 0 or cost < low:
        low = cost

print("lowest cost:", low)
```

OUTPUT (SAMPLE)

```
Enter number of vehicles: 2
Enter distance: 100
Enter reserve fuel: 2
Enter mileage: 20
Enter fuel price: 100
fuel needed: 7.0
journey cost: 700.0
Enter mileage: 25
```

```
Enter fuel price: 100
fuel needed: 6.0
journey cost: 600.0
lowest cost: 600.0
```

PROGRAM 4: Product Billing with Discount and Tax

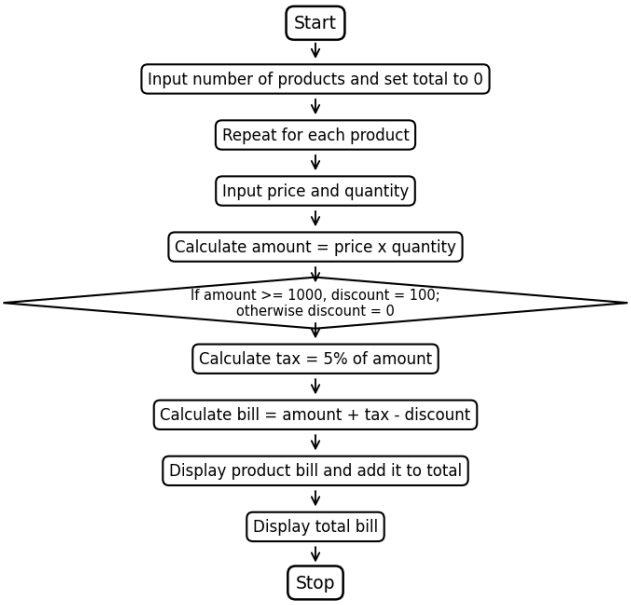
AIM

Calculate each product bill using price and quantity, apply a fixed discount when the amount is at least 1000, add 5% tax and calculate the total bill.

PROCEDURE

1. Input number of products and set total to 0
2. Repeat for each product
3. Input price and quantity
4. Calculate amount = price x quantity
5. If amount >= 1000, discount = 100; otherwise discount = 0
6. Calculate tax = 5% of amount
7. Calculate bill = amount + tax - discount
8. Display product bill and add it to total
9. Display total bill

FLOWCHART



PROGRAM

```
n = int(input("Enter number of products: "))
total = 0

for i in range(n):
    price = int(input("Enter price: "))
    quantity = int(input("Enter quantity: "))

    amount = price * quantity

    if amount >= 1000:
        discount = 100
    else:
        discount = 0

    tax = amount * 0.05
    bill = amount + tax - discount

    print("product bill:", bill)

    total = total + bill

print("total bill:", total)
```

OUTPUT (SAMPLE)

```
Enter number of products: 2
Enter price: 500
Enter quantity: 2
product bill: 950.0
Enter price: 200
Enter quantity: 3
```

```
product bill: 630.0
total bill: 1580.0
```

PROGRAM 5: Production Cost and Profit Target

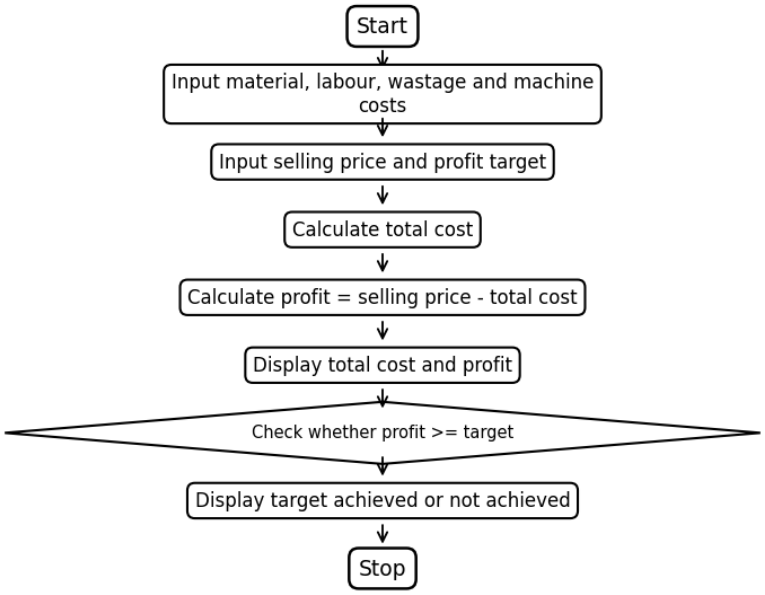
AIM

Find total production cost and profit, then check whether the profit reaches the given target.

PROCEDURE

1. Input material, labour, wastage and machine costs
2. Input selling price and profit target
3. Calculate total cost
4. Calculate profit = selling price - total cost
5. Display total cost and profit
6. Check whether profit >= target
7. Display target achieved or not achieved

FLOWCHART



PROGRAM

```
material = int(input("Enter material cost: "))
labour = int(input("Enter labour charges: "))
wastage = int(input("Enter wastage cost: "))
machine = int(input("Enter machine cost: "))
selling = int(input("Enter selling price: "))
target = int(input("Enter profit target: "))

cost = material + labour + wastage + machine
profit = selling - cost

print("total cost:", cost)
print("profit:", profit)

if profit >= target:
    print("profit target achieved")
else:
    print("profit target not achieved")
```

OUTPUT (SAMPLE)

```
Enter material cost: 500
Enter labour charges: 200
Enter wastage cost: 50
Enter machine cost: 100
Enter selling price: 1000
Enter profit target: 100
total cost: 850
profit: 150
profit target achieved
```

PROGRAM 6: Event Budget and Minimum Registrations

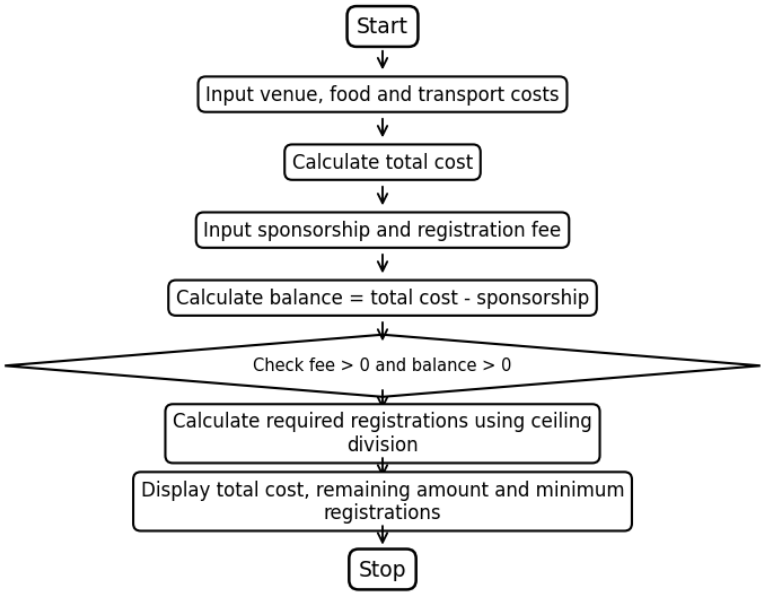
AIM

Calculate total event cost and remaining amount after sponsorship. Find the minimum registrations needed to cover the remaining amount using the registration fee.

PROCEDURE

1. Input venue, food and transport costs
2. Calculate total cost
3. Input sponsorship and registration fee
4. Calculate balance = total cost - sponsorship
5. Check fee > 0 and balance > 0
6. Calculate required registrations using ceiling division
7. Display total cost, remaining amount and minimum registrations

FLOWCHART



PROGRAM

```
venue = int(input("Enter venue cost: "))
food = int(input("Enter food cost: "))
transport = int(input("Enter transport cost: "))

total = venue + food + transport

sponsor = int(input("Enter sponsorship amount: "))
fee = int(input("Enter registration fee per person: "))

balance = total - sponsor

print("total cost:", total)
print("remaining amount:", balance)

if fee > 0 and balance > 0:
    count = balance // fee
    if balance % fee != 0:
        count = count + 1
else:
    count = 0

print("minimum registrations:", count)
```

OUTPUT (SAMPLE)

```
Enter venue cost: 5000
Enter food cost: 3000
Enter transport cost: 2000
Enter sponsorship amount: 4000
Enter registration fee per person: 300
total cost: 10000
remaining amount: 6000
minimum registrations: 20
```

PROGRAM 7: Student Attendance Eligibility

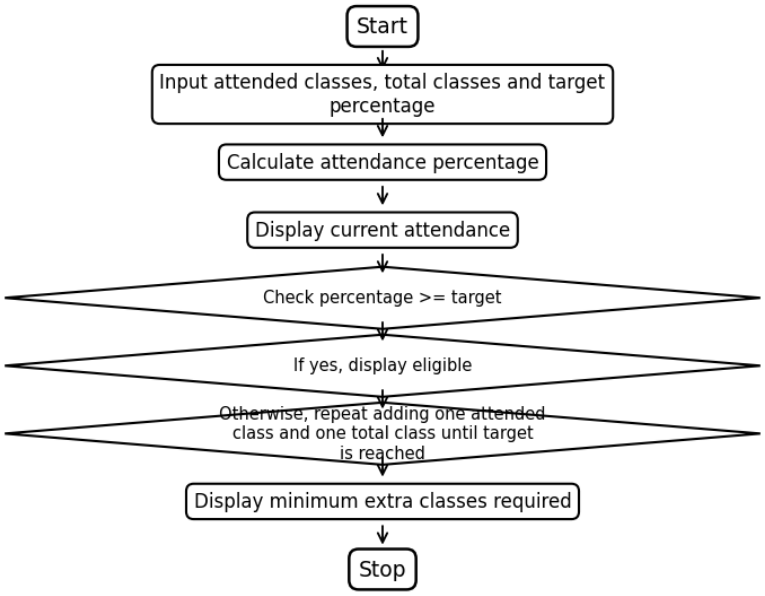
AIM

Calculate current attendance percentage and determine eligibility. If attendance is below the target, calculate the minimum additional classes needed to reach it.

PROCEDURE

1. Input attended classes, total classes and target percentage
2. Calculate attendance percentage
3. Display current attendance
4. Check percentage >= target
5. If yes, display eligible
6. Otherwise, repeat adding one attended class and one total class until target is reached
7. Display minimum extra classes required

FLOWCHART



PROGRAM

```
attended = int(input("Enter attended classes: "))
total = int(input("Enter total classes: "))
target = int(input("Enter required attendance: "))

percentage = attended / total * 100
print("current attendance:", percentage)

if percentage >= target:
    print("eligible")
else:
    extra = 0
    while percentage < target:
        attended = attended + 1
        total = total + 1
        extra = extra + 1
        percentage = attended / total * 100

    print("minimum extra classes required:", extra)
```

OUTPUT (SAMPLE)

```
Enter attended classes: 40
Enter total classes: 50
Enter required attendance: 75
current attendance: 80.0
eligible
```

PROGRAM 8: Bank Deposit and Withdrawal Transactions

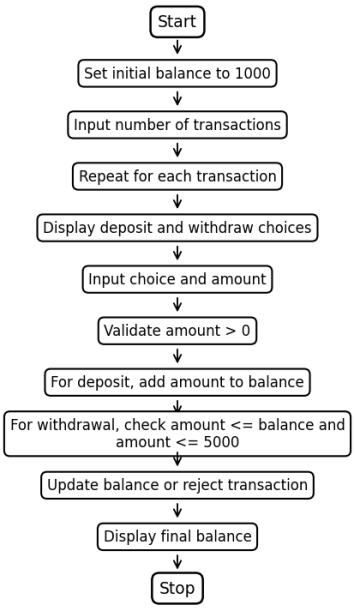
AIM

Process a series of deposit and withdrawal transactions, validate amounts and withdrawal limits, and display the final balance.

PROCEDURE

1. Set initial balance to 1000
2. Input number of transactions
3. Repeat for each transaction
4. Display deposit and withdraw choices
5. Input choice and amount
6. Validate amount > 0
7. For deposit, add amount to balance
8. For withdrawal, check amount <= balance and amount <= 5000
9. Update balance or reject transaction
10. Display final balance

FLOWCHART



PROGRAM

```
balance = 1000
n = int(input("Enter number of transactions: "))

for i in range(n):
    print("1. deposit")
    print("2. withdraw")
    choice = int(input("Enter choice: "))
    amount = int(input("Enter amount: "))

    if amount <= 0:
        print("invalid amount")
    elif choice == 1:
        balance = balance + amount
        print("deposit successful")
    elif choice == 2:
        if amount <= balance and amount <= 5000:
            balance = balance - amount
            print("withdrawal successful")
        else:
            print("transaction rejected")
    else:
        print("invalid choice")

print("final balance:", balance)
```

OUTPUT (SAMPLE)

```
Enter number of transactions: 3
1. deposit
2. withdraw
Enter choice: 1
```

```
Enter amount: 500
deposit successful
1. deposit
2. withdraw
Enter choice: 2
Enter amount: 200
withdrawal successful
1. deposit
2. withdraw
Enter choice: 2
Enter amount: 2000
transaction rejected
final balance: 1300
```

PROGRAM 9: Parking Fee Calculation

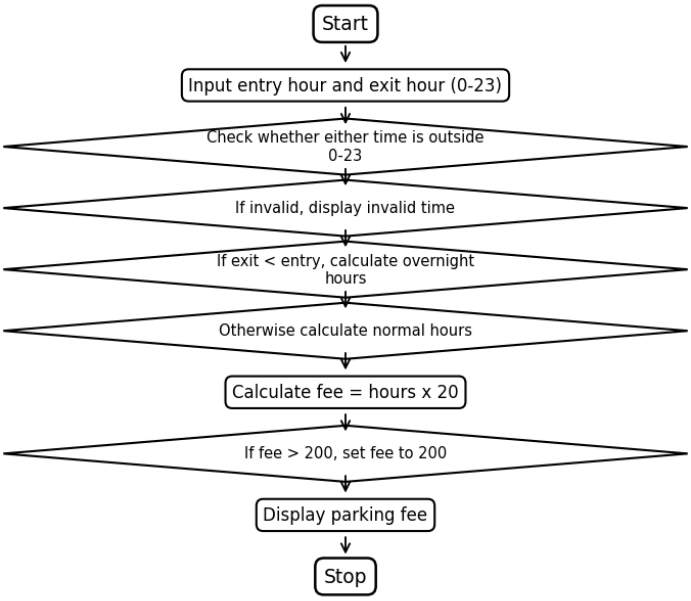
AIM

Calculate parking duration and fee from entry and exit hours. Handle overnight parking and cap the fee at 200.

PROCEDURE

1. Input entry hour and exit hour (0-23)
2. Check whether either time is outside 0-23
3. If invalid, display invalid time
4. If exit < entry, calculate overnight hours
5. Otherwise calculate normal hours
6. Calculate fee = hours x 20
7. If fee > 200, set fee to 200
8. Display parking fee

FLOWCHART



PROGRAM

```
entry = int(input("Enter entry hour (0-23): "))
exit = int(input("Enter exit hour (0-23): "))

if entry < 0 or entry > 23 or exit < 0 or exit > 23:
    print("invalid time")
elif exit < entry:
    print("overnight parking")
    hours = 24 - entry + exit
    fee = hours * 20
    if fee > 200:
        fee = 200
    print("parking fee:", fee)
else:
    hours = exit - entry
    fee = hours * 20

    if fee > 200:
        fee = 200

    print("parking fee:", fee)
```

OUTPUT (SAMPLE)

```
Enter entry hour (0-23): 22
Enter exit hour (0-23): 2
overnight parking
parking fee: 80
```

PROGRAM 10: Hospital Billing and Patient Classification

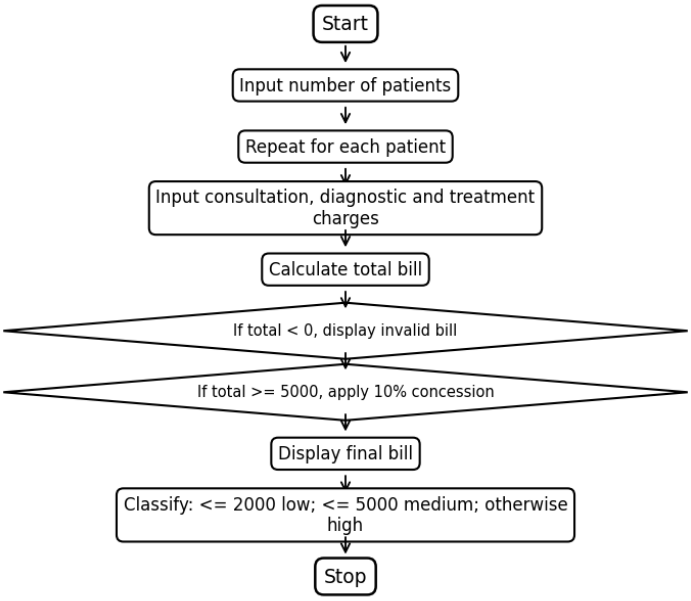
AIM

Calculate each patient's bill from consultation, diagnostic and treatment charges. Reject negative totals, apply a 10% concession for bills of at least 5000, and classify the final bill as low, medium or high.

PROCEDURE

1. Input number of patients
2. Repeat for each patient
3. Input consultation, diagnostic and treatment charges
4. Calculate total bill
5. If total < 0, display invalid bill
6. If total >= 5000, apply 10% concession
7. Display final bill
8. Classify: <= 2000 low; <= 5000 medium; otherwise high

FLOWCHART



PROGRAM

```
n = int(input("patients: "))

for i in range(n):
    c = int(input("consultation: "))
    d = int(input("diagnostic: "))
    t = int(input("treatment: "))
    total = c + d + t

    if total < 0:
        print("invalid bill")
    else:
        if total >= 5000:
            total = total * 0.9

        print("bill:", total)

        if total <= 2000:
            print("low")
        elif total <= 5000:
            print("medium")
        else:
            print("high")
```

OUTPUT (SAMPLE)

```
patients: 2
consultation: 500
diagnostic: 300
treatment: 700
bill: 1500
low
consultation: 2000
```

```
diagnostic: 1500
treatment: 2500
bill: 5400.0
high
```